Laplace transform

$$F(s) = \mathcal{L}[f(t)] = \int_{0^-}^{\infty} f(t)\, e^{-st}\, dt$$

one sided for $t > 0$

$e^{-at} \leftrightarrow \frac{1}{s+a}$ $\quad \sin\omega t \leftrightarrow \frac{\omega}{s^2+\omega^2}$

$\frac{df}{dt} \leftrightarrow sF(s) - f(0^-)$ $\quad \cos\omega t \leftrightarrow \frac{s}{s^2+\omega^2}$

Table 14.2-1 Laplace Transform Pairs

$f(t)$ for $t > 0$	$F(s) = \mathcal{L}[f(t)u(t)]$
$\delta(t)$	1
$u(t)$ 1	$\frac{1}{s}$
e^{-at}	$\frac{1}{s+a}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
$e^{-at}t^n$	$\frac{n!}{(s+a)^{n+1}}$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{-at}\sin(\omega t)$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos(\omega t)$	$\frac{s+a}{(s+a)^2+\omega^2}$

Table 14.2-2 Laplace Transform Properties

PROPERTY	$f(t),\ t > 0$	$F(s) = \mathcal{L}[f(t)u(t)]$
Linearity	$a_1 f_1(t) + a_2 f_2(t)$	$a_1F_1(s) + a_2F_2(s)$
Time scaling	$f(at)$, where $a > 0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Time integration	$\int_0^t f(\tau)d\tau$	$\frac{1}{s}F(s)$
Time differentiation	$\frac{df(t)}{dt}$	$sF(s) - f(0^-)$
	$\frac{d^2f(t)}{dt^2}$	$s^2F(s) - \left(sf(0^-) + \frac{df(0^-)}{dt}\right)$
	$\frac{d^nf(t)}{dt^n}$	$s^nF(s) - \sum_{k=1}^{n} s^{n-k}\frac{d^{k-1}f(0^-)}{dt^{k-1}}$
Time shift	$f(t-a)u(t-a)$	$e^{-as}F(s)$
Frequency shift	$e^{-at}f(t)$	$F(s+a)$
Time convolution	$f_1(t)*f_2(t) = \int_0^t f_1(\tau)f_2(t-\tau)d\tau$	$F_1(s)F_2(s)$
Frequency integration	$\frac{f(t)}{t}$	$\int_s^\infty F(\lambda)d\lambda$
Frequency differentiation	$tf(t)$	$-\frac{dF(s)}{ds}$
Initial value	$f(0^+)$	$\lim_{s\to\infty} sF(s)$
Final value	$f(\infty)$	$\lim_{s\to 0} sF(s)$

$A\cos(\omega t+\theta) = (A\cos\theta)\cos(\omega t) - (A\sin\theta)\sin(\omega t)$

Inverse Laplace Transform: Simple Real poles.

$$F(s) = \frac{N(s)}{D(s)} = \frac{b_m s^m + b_{m-1}s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

$$= \frac{R_1}{s-p_1} + \frac{R_2}{s-p_2} + \cdots + \frac{R_i}{s-p_i} + \cdots + \frac{R_n}{s-p_n}$$

$$R_i = (s-p_i)F(s)\big|_{s=p_i}$$

Repeated poles

$$F(s) = \frac{R_1}{s-p_1} + \frac{R_2}{(s-p_1)^2} + \cdots + \frac{R_q}{(s-p_1)^q} + \frac{R_{q+1}}{s-p_{q+1}} + \cdots + \frac{R_n}{s-p_n}$$

$$R_{q-k} = \frac{1}{k!}\left[\frac{d^k}{ds^k}(s-p_1)^q F(s)\right]\Big|_{s=p_1} \quad \text{for } k = q-1,\ q-2,\ \ldots,\ 2,\ 1,\ 0$$

kouncer dereceden türev alıyoruz

t is,

$$R_1 = \frac{1}{(q-1)!}\left[\frac{d^{q-1}}{ds^{q-1}}(s-p_1)^q F(s)\right]\Big|_{s=p_1},$$

$$R_2 = \frac{1}{(q-2)!}\left[\frac{d^{q-2}}{ds^{q-2}}(s-p_1)^q F(s)\right]\Big|_{s=p_1}, \ldots$$

$$R_q = [(s-p_1)^q F(s)]|_{s=p_1}$$

Improper Rational function

* payın derecesi büyükse polinom bölmesi yapılır

Circuit Analysis Using Impedance and Initial Conditions

$v(t) = i(t)R \leftrightarrow V(s) = I(s)R$

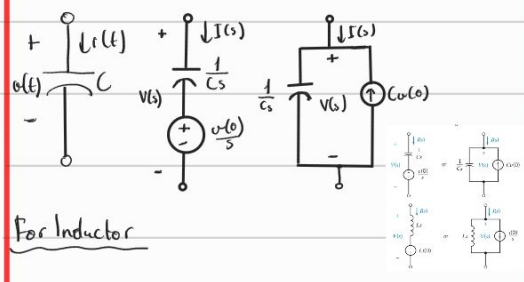
$\Rightarrow Z(s) = \frac{V(s)}{I(s)}$

For capacitor

$v(t) = \frac{1}{C}\int_0^t i(z)dz + v(0) \leftrightarrow V(s) = \frac{1}{Cs}I(s) + \frac{v(0)}{s}$

$\Rightarrow Z_C(s) = \frac{1}{Cs}$

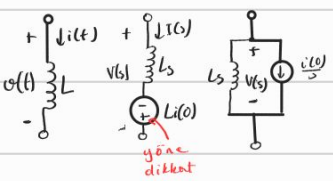
and $I(s) = CsV(s) - Cv(0)$



For Inductor

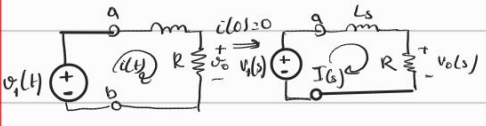
$v(t) = L\frac{d}{dt}i(t) \leftrightarrow V(s) = LsI(s) - Li(0)$

$Z_L(s) = Ls$ and $I(s) = \frac{1}{Ls}V(s) + \frac{i(0)}{s}$



Transfer function and impedance

$H(s) = \frac{V_o(s)}{V_i(s)}$



$V_o(s) = H(s)V_i(s)$

when the input is a unit step function

$V_i(s) = \mathcal{L}[u(t)] = \frac{1}{s}$

$V_o(s) = \frac{H(s)}{s}$

$\Rightarrow$ Step response $= \mathcal{L}^{-1}\left[\frac{H(s)}{s}\right]$

when the input is an impulse function

$v_i(t) = \delta(t) \leftrightarrow V_i(s) = \mathcal{L}[\delta(t)] = 1$

$V_o(s) = H(s)$

Impulse response $= \mathcal{L}^{-1}[H(s)]$

$H(s) =$ Voltage gain $= \frac{V_o(s)}{V_i(s)}$

$H(s) =$ current gain $= \frac{I_o(s)}{I_i(s)}$

$H(s) =$ Impedance $= \frac{V(s)}{I(s)}$

$H(s) =$ Admittance $= \frac{I(s)}{V(s)}$

The Fourier Series

Trigonometric Fourier series

$$f(t) = f(t+nT)$$

$$f(t) = \underbrace{a_0}_{\text{dc}} + \sum_{n=1}^{\infty} \underbrace{(a_n \cos n\omega_0 t + b \sin n\omega_0 t)}_{\text{ac}}$$

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\,dt$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t\,dt$$

Asalternative

$$f(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \phi_n)$$

$$A_n = \sqrt{a_n^2 + b_n^2}, \quad \phi_n = -\tan^{-1}\frac{b_n}{a_n}$$

important integrals

$$\int \cos at\,dt = \frac{1}{a}\sin at$$

$$\int \sin at\,dt = -\frac{1}{a}\cos at$$

$$\int t\cos at\,dt = \frac{1}{a^2}\cos at + \frac{1}{a}t\sin at$$

$$\int t\sin at\,dt = \frac{1}{a^2}\sin at - \frac{1}{a}t\cos at$$

Symetry Considerations

Even Symmetry (Genap)

$$f(t) = f(-t)$$

$$a_0 = \frac{2}{T}\int_0^{T/2} f(t)\,dt$$

$$a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos n\omega_0 t\,dt \qquad b_n = 0$$

Odd Symmetry (teksimetri)

$$f(-t) = -f(t) \qquad \int_{-T/2}^{T/2} f_0(t)\,dt = 0$$

$$a_0 = 0 \qquad a_n = 0$$

$$b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin n\omega_0 t\,dt$$

Halfwave Symmetry

$$f\left(t - \frac{T}{2}\right) = -f(t)$$

$$a_0 = 0$$

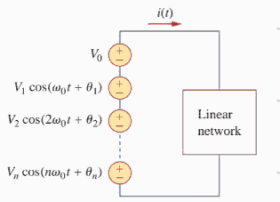
$$a_n = \begin{cases} \frac{4}{T}\int_0^{T/2} f(t)\cos n\omega_0 t\,dt, & \text{for n odd} \\ 0, & \text{for n even} \end{cases}$$

$$b_n = \begin{cases} \frac{4}{T}\int_0^{T/2} f(t)\sin n\omega_0 t\,dt, & \text{for n odd} \\ 0 & \text{for n even} \end{cases}$$

Circuit Aplications

$$v(t) = V_0 + \sum_{n=1}^{\infty} V_n \cos(n\omega_0 t + \theta_n)$$

$v(t)$ consists of two parts: the dc component V_0 and the ac component $V_n = |V_n| \angle \theta_n$ with several harmonics



So $i(t) = I_0 + \sum_{n=1}^{\infty} |I_n| \cos(n\omega_0 t + \psi_n)$

Average Power and RMS Values

$$v(t) = V_{dc} + \sum_{n=1}^{\infty} V_n \cos(n\omega_0 t - \theta_n)$$

$$i(t) = I_{dc} + \sum_{n=1}^{\infty} I_n \cos(n\omega_0 t - \phi_n)$$

$$P = V_{dc} I_{dc} + \frac{1}{2}\sum_{n=1}^{\infty} V_n I_n \cos(\theta_n - \phi_n)$$

$$\boxed{F_{rms} = \sqrt{a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2)}}$$

Frequency Response

$$H(\omega) = \frac{Y(\omega)}{X(\omega)}$$

The decibal Scale

1. $\log P_1 P_2 = \log P_1 + \log P_2$

2. $\log P_1/P_2 = \log P_1 - \log P_2$

3. $\log P^n = n \log P$

4. $\log 1 = 0$

G = Number of bels $= \log_{10} \frac{P_2}{P_1}$

$G_{dB} = 10 \log_{10} \frac{P_2}{P_1}$

$G_{dB} = 20 \log_{10} \frac{V_2}{V_1}$

$G_{dB} = 20 \log_{10} \frac{I_2}{I_1}$

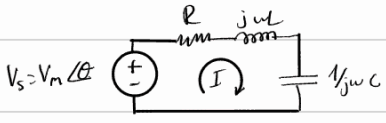
$H = |H| \angle \Phi = H e^{j\theta}$

$\ln H = \ln|H| + \ln e^{j\theta} = \ln|H| + j\theta$

$H_{dB} = 20 \log_{10} H$

Factor	Magnitude	Phase
K	$20 \log_{10} K$	$0°$
$(j\omega)^N$	$20N$ dB/decade through $(1, 0)$	$90N°$
$\frac{1}{(j\omega)^N}$	$-20N$ dB/decade through $(1, 0)$	$-90N°$
$\left(1 + \frac{j\omega}{z}\right)^N$	$20N$ dB/decade through $(z, 0)$	$0°$ for $\omega \le \frac{z}{10}$; $90N°$ for $\omega \ge 10z$
$\frac{1}{(1 + j\omega/p)^N}$	$-20N$ dB/decade through $(p, 0)$	$0°$ for $\omega \le \frac{p}{10}$; $-90N°$ for $\omega \ge 10p$
$\left[1 + \frac{2j\omega\zeta}{\omega_n} + \left(\frac{j\omega}{\omega_n}\right)^2\right]^N$	$40N$ dB/decade through $(\omega_n, 0)$	$0°$ for $\omega \le \frac{\omega_n}{10}$; $180N°$ for $\omega \ge 10\omega_n$
$\frac{1}{[1 + 2j\omega\zeta/\omega_k + (j\omega/\omega_k)^2]^N}$	$-40N$ dB/decade through $(\omega_k, 0)$	$0°$ for $\omega \le \frac{\omega_k}{10}$; $-180N°$ for $\omega \ge 10\omega_k$

Series Resonance



$$Z = H(\omega) = \frac{V_s}{I} = R + j\omega L + \frac{1}{j\omega C}$$

$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$Im(Z) = \omega L - \frac{1}{\omega C} = 0$$

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}}$$

Since $\omega_0 = 2\pi f_0$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

The frequency response of the circuit's current magnitude

$$I = |I| = \frac{V_m}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

$$P(\omega) = \frac{1}{2} I^2 R \qquad P(\omega_0) = \frac{1}{2}\frac{V_m^2}{R}$$

$$P(\omega_1) = P(\omega_2) = \frac{(V_m/\sqrt{2})^2}{2R} = \frac{V_m^2}{4R}$$ ← half power

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2}\,R$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}}$$

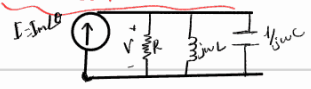
$$Q = 2\pi \frac{\frac{1}{2} L I^2}{\frac{1}{2} I^2 R (1/f_0)} = \frac{2\pi f_0 L}{R}$$

$$\boxed{Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}}$$

$$\boxed{\text{Bandwith} = \frac{R}{L} = \frac{\omega_0}{Q}}$$

$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \quad \omega_2 \simeq \omega_0 + \frac{B}{2}$$

Parallel Resonance



$$Y = H(\omega) = \frac{I}{V} = \frac{1}{R} + j\omega C + \frac{1}{j\omega L}$$

$$Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

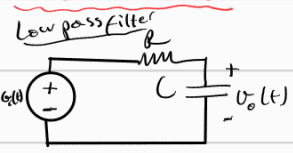
$$\omega_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$\boxed{B = \omega_2 - \omega_1 = \frac{1}{RC}} \qquad \boxed{Q = \frac{\omega_0}{B} = \omega_0 RC = \frac{R}{\omega_0 L}}$$

Passive Filters

Low pass filter



$$H(\omega) = \frac{V_o}{V_i} = \frac{1/j\omega C}{R + 1/j\omega C}$$

$$H(\omega) = \frac{1}{1 + j\omega RC}$$

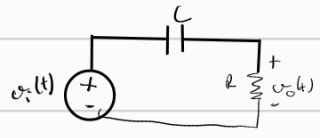
* $H(0) = 1$, $H(\infty) = 0$

ω_c → cutoff frequency

$$H(\omega_c) = \frac{1}{\sqrt{1 + \omega_c^2 R^2 C^2}} = \frac{1}{\sqrt{2}}$$

$$\boxed{\omega_c = \frac{1}{RC}}$$

Highpass filter

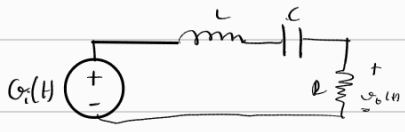


$$H(\omega) = \frac{V_o}{V_i} = \frac{R}{R + 1/j\omega C}$$

$$H(\omega) = \frac{j\omega RC}{1 + j\omega RC}$$

* $H(0) = 0$, $H(\infty) = 1 \Rightarrow \omega_c = \frac{1}{RC}$

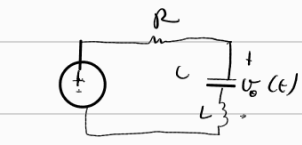
Band pass filter



$$H(\omega) = \frac{V_o}{V_i} = \frac{R}{R + j(\omega L - 1/\omega C)}$$

$$\omega_o = \frac{1}{\sqrt{LC}}$$

Bandstop filt.



$$H(\omega) = \frac{V_o}{V_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)}$$

Notice that $H(0) = 1$, $H(\infty) = 1$

and $\omega_o = \frac{1}{\sqrt{LC}}$ //